The Effect of Color Codes on Temporal Order Judgements

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Abstract

The understanding of the causal relationship between events or their temporal order in relation to each other implies the experience of unravelling the meanings behind the form in which they are presented in order to draw appropriate conclusions. Previously, the topic of how temporal words influence judgments regarding the temporal sequence of events has been severely studied. It appears to be the case, that the 'frame' or the form of the events' presentation plays an important role in how the process of their perception is handled and how the judgements about them are derived. However, there is a lack of research regarding how additional stimuli can affect and evaluate the event's perception and inference.

The present study seeks to address this issue by examining the impact of visual colour cues on the subjective alignment of perception and inferences of temporal order judgments by measuring event-related potentials (ERP) components triggered by the words "before" and "after", which is conducted with the help of electroencephalography (EEG) system. Within the present paper, I suggest an experiment design that could be further improved and put into practice. According to the experiment, participants are presented with a number of two-clause sentences separated by a comma and containing temporal words "before" and "after" at varying positions in the different sentences, while a number of clauses in the sentences are colour-coded. The results comparison is suggested to be based on the derived reaction time data and correctness of temporal order judgements to the non-coloured and colour-coded stimuli within each participant, in a manner that every subject is exposed to the non-coloured and colour-coded stimuli to a similar extent.

The study hypothesises that additional visual information will lead to better temporal order judgments represented in the correctness of provided answers to the temporal order-related questions, as well as faster reaction times and greater comprehension accuracy. It is further proposed that the response to coloured stimuli will record fewer negative ERPs due to lower processing costs for making temporal order judgments. It is assumed, as the subjects can see a clearer separation between the clauses and temporal words due to the colour coding, they will focus more on the temporal connectives rather than on the sentence narrative, leading to faster reactions and more accurate responses.

Keywords: temporal order judgement, ERP, EEG, connectives, visual colour stimuli

1 Introduction

The subject of temporal order judgements has been the interest of many studies over the last decades. Most studies on temporal order judgments had their focus on investigating how individuals perceive and interpret the order of events based on various contextual cues, including temporal words such as "before," "after," and "during." These studies have examined the effects of different factors, such as the presence or absence of contextual information, the complexity of the stimuli, and the level of cognitive load on temporal order judgments. The questions of interest were mainly about the relation of temporal order to working memory [1, 2, 3], world knowledge and in terms of truth-value [2] or sentence structure conditions [4, 3]. In the present study, I draw upon the findings and content of the previous experiments.

Despite the lack of scientific research in this regard, previous studies have shown that temporal order judgments can be influenced by a range of factors, including the modality of presented stimuli (visual, auditory [7, 8]), an addition of choice [5], as well as attention, memory, and even the sense of ownership [9].

In this regard, EEG and ERP techniques were previously used to measure brain activity associated with temporal order judgments [1, 3]. These techniques allowed researchers to investigate the neural mechanisms underlying temporal order judgments and to examine how different brain regions and networks are involved in this process.

First and foremost, the most significant results within the framework of the present paper were obtained earlier in the study by Münte et al. [1], where the principles of how the brain processes the relationship between temporal terms (e.g., "before" and "after") and the conceptual order of events were studied. The stimuli sentences were presented as two-clause sentences (initial subordinate clause and subsequent main clause), with each of the clauses describing a particular event not logically or causally related to one another, as in the following example: *"'Before/After' the psychologist submitted the article, the journal changed its policy"* [1]. Referring to the real-world knowledge of time and causality of events, as well as the linguistic knowledge of how would these statements be mapped onto expressions of time, authors expressed the concern that the first word might subconsciously lead readers towards the opinion about the presence of the non-arbitrary relationship between the clause-statements. Authors also expected to see a tendency towards higher mental costs during the processing of "before" sentences compared to "after" sentences. Münte et al. explain this assumption by participants' implicit need to "mentally rearrange" the structure of a given sentence to fit it in the natural temporal order of occurring events as they would happen in the real world. The study used event-related potentials (ERPs) in order to measure the neural activity of participants during the trials, where they read sentences exposing temporal terms in a way that conflicted with the actual order of events in the sentence (e.g. the event that according to the narrative is happened later in time is placed at the beginning of the sentence, and vice versa). The authors hypothesized that processing these conflicting temporal terms would require additional cognitive resources and result in a delay or disruption of the neural response.

Overall, the results of the study showed that the processing of conflicting temporal terms did indeed result in a delay in the ERP response, suggesting that additional cognitive resources were required to resolve the conflict. Considering the correlation of the derived results to working memory skills, researchers have found that compared to the low working memory scores, among the results of subjects with the high working memory measurements, signs of greater negativity within 300 ms for "before"-sentences were found compared to "after"-sentences. ERP recordings elicited for "after"-sentences were more positive than for "before"-sentences. These findings indicated the difficulty of comprehension of the "after"-sentences rather than "before"-sentences.

Similarly, the paper of Zwaan et al. [6] states that examination regarding the subject of temporal order also requires considering the way the brain naturally perceives the timeline of occurring events. In particular, any deviations in the chronological order seem to add difficulties during the events' mental processing. Zwaan et al. built their study on the testing of "iconicity assumption in language comprehension." Typically, people tend to comprehend events in chronological order, i.e., as they would occur naturally. Thus, it states the difference in comprehension of "before" and "after" sentences in favour of the latter - participants' reaction time is higher, and they are less accurate while reading sentences starting with "before" in comparison to sentences having "after" in the beginning. In the study, temporal order two-clause sentences were structured in a similar way as in the mentioned study of Münte et al., using "before" and "after" constructions. This study showed a specific difference in the comprehension of temporal order sentences, particularly the evidence favoring a strong-iconicity assumption (comprehenders process events as if they occurred in chronological order, additionally assuming their contiguity) [6].

Respectively, the evidence of the relevance of the chronological order of events presented in the sentences was previously derived by Mandler [4]. Mandler found out that participants' reading time scores were significantly higher for sentences with violated chronological order (except the situation where clauses' events had causation between each other).

These ideas were combined and realized in the research of Politzer-Ahles et al. [3]. The study was built as a 2x2 design, using temporal connectives ("before" vs. "after") and changing sentences' structure (sentence-initial temporal clause vs. sentence-final temporal clause), placing the trigger word ("before" or "after") either at the beginning of the sentence or the beginning of the second clause right after the comma separating clauses. For the latter case, build as in the following example: *"Jens submitted his resignation, before/after he received a promotion"*. The experiment's main aim was to examine the effect of structure concerning the comprehension of temporal order sentences. Authors used comprehension questions about the content of the sentences that followed a third of the items to probe the extent of the correctness of understanding the order in which the sentence's events occurred. The EEG recording detected sustained negativity in ERPs for "before"-clauses in sentence-initial temporal clauses than for "after"-clauses. Considering the sentence-final temporal clauses, the situation was the right opposite, "after"-clauses elicited more negative ERPs than "before"-clauses. The results also contained evidence in favor of the traditional account of the order-of-mention hypothesis: the difference between the order in which the events are presented in the sentences and the order in which they would naturally occur on the timeline adds to mental processing costs that one can observe during the analysis of temporal clauses comprehension [3].

2 The Focus of the Present Study

There have been several paradigms that investigate the principles of temporal order information processing in different modalities, e.g. visual and auditory [7, 8]. By examining these modalities in different variations in a combination with the temporal order sentences, one might reveal and measure the effects of such modalities on the perception of the temporal order judgements. Within the present study, examining the role of colour coding in relation to the comprehension of the stimuli and the narrative of the presented sentences is of special interest. I will focus on presenting the participants the sets of 'before-' and 'after-' based sentences in two different variations colour-wise - non-coloured clause sentences and colour-coded clause sentences - in order to examine whether there is a difference in the patterns of perception of these stimuli. Thus, apart from processing costs that 'before' and 'after' containing sentences cause alone, I aim to investigate and analyze how external other sensory modalities' information helps to derive temporal order judgments. In earlier studies, it was found that attention facilitates visual stimuli identification [5]. In the present study, based on the ideas of Politzer-Ahles et al. [3], I determine how this influences subsequent judgments about various stimuli' features concerning their temporal order.

My aim within the present study is to provide an experiment design where apart from the temporal order-related sentences, visual information is also assigned to the events in the 'before-' and 'after-' based sentence stimuli and contributed to the subjects in a way that presumably helps to derive temporal order judgments of the said events more accurately and in a less amount of time.

EEG recording is done upon presentation of the stimuli. The acquired EEG responses are needed in order to understand the effect of the visual component of stimuli on temporal order judgments. ERP recordings at stimulus onset would be beneficial for examining the participants' perceptual decisions on the temporal order sentences with or without the visual effect. It is expected, that visual colour stimulus helps to separate the event clauses from the temporal trigger words, thus, less attention, effort and time would be paid to the comprehension of the narrative of the sentence and analyzing of the potential causality between events.

Thus, I hypothesise the following: (1) there are more negative ERPs to temporal connectives if the participants are presented with visual colour stimuli when compared with the non-coloured stimuli; (2) the percentage of the correctness of answers to the event- and temporal-order-related questions will be higher. Accordingly, the reaction time will be faster for the experimental trials (with colour-coding stimuli) than for the control trials (non-coloured stimuli).

2.1 Design of the Experiment

For the experiment, I propose a 2x2x2 factor design to study the effects of Connective (before vs. after) x Structure (sentence-initial vs. sentence-final temporal clause) x Visual Prime (colour coded vs. non-coloured). All participants are presented with both experimental and control trials. I aim to use a standard EEG procedure of electrode placements of 32 channels according to the 10-20 system. At the beginning of each trial, participants are presented with a fixation point in the middle of the screen for 650 ms, after that, either colour-coded or non-coloured randomly chosen stimulus sentence follows that fashions events in a prescribed temporal order. EEG recording is done upon the presentation of the stimulus. The recorded ERP shows the reaction of the participants on triggers - temporal order words - 'before' / 'after' as they occur in the presented sentence. This procedure is kept standard for each participant.

3 Method

3.1 Participants

It is recommended, that twenty University students between the ages of 20 and 29 be required to participate in this study. All participants are native English speakers and are right-handed. Any participants with neurological or psychiatric disorders will be excluded from the study.

3.2 Materials

For each participant, two sets of 12 sentences are used (12 colour-coded and 12 non-coloured) that are in total 24 sentence stimuli. The set of colour-coded sentences appears to vary in colour for each clause. That is done so no conclusions are derived about the connection between the colour and the temporal order of the events to which one or another colour is assigned. Only two colours are used for the colour coding - blue and red. No repeating of the colours occurs within one sentence. Thus, all the possible variations are blue-red, red-blue, and black-black (i.e. non-coloured). The sentences are presented in a way, that the amount of non-coloured stimuli and colour-coded stimuli are equal.

The experiment itself is structured into two blocks with two sets of 12 sentences each, and in each block, the sentences of the 'before' and 'after'-sets are presented in fully random order. In the first block, the focus lies on sentence-initial temporal trigger stimuli starting with the temporal clauses 'before' or 'after'. In the second block, the focus lies on sentence-final temporal trigger stimuli, where the subordinate clauses start with the temporal clauses 'before' or 'after'. In Tables 1. and 2., one can get familiar with a set of example sentences.

	'before'	'after'
Sentence-initial	<u>Before</u> I came in, it started to rain	<u>After</u> I came in, it started to rain
Sentence-final	I came in, <u>before</u> it started to rain	I came in, <u>after</u> it started to rain

Table 1: Example Sentence Stimuli

Colour-coded	Non-coloured
Before I came in, it started to rain	Before I came in, it started to rain

Table 2: Example Colour Stimuli

3.3 Procedure

The participants are seated in an electrically-shielded and sound-attenuated booth in front of a computer in the experiment. They read a total of 24 stimulus sentences (in black 20-point font on a white background) and answer the raised at the beginning of the experiment question that remains unchanged for every trial, using a computer keyboard, as the electroencephalogram (EEG) is recording during the whole process.

Upon fixating on a small cross in the monitor's centre (650ms) followed by a blank space (1s), the participants will see the presented stimulus, one per trial.

There is no time limit for the stimulus presentation and answer to the question, however, the reaction time is recorded as soon as the stimulus is onset.

All participants are instructed at the beginning of the experiment to respond to the presented question by pressing the respective key combinations on the keyboard: Key 'S' indicating 'Clause 1' and key 'K' indicating 'Clause 2'. The raised question is the following: *"Please pick the clause that happened EARLIER"*. The question remains the same for each trial in two blocks. The changing colour-coding for the clauses is also done to balance out any potential effects of including only right-handed participants in the study, so there is no causality derived between the colour and keys 'S' and 'K'.

4 Expected Results

When it comes to the results of the present study, I expect to see the presence of any evidence in favour of the stated hypotheses, which are the following:

Concerning our first hypothesis:

(1) There are more negative ERPs to temporal connectives if the participants are presented with visual colour stimuli when compared with the non-coloured stimuli.

I suppose that there will be greater negativity regarding ERPs to the colour-coded stimuli in comparison to the ERPs derived when participants are exposed to the non-coloured stimuli, as the colour-coded stimulus is expected the relieve the cognitive tension of and lessen the efforts spent on determining the temporal order of presented events within the sentence.
Regarding the temporal order judgements, while comparing reactions to the stimuli across connectivity ('after' vs. 'before'; sentence-initial vs. sentence-final), I expect to see greater negativity ERPs for "before"-clauses when compared to "after"-clauses (more positive) in sentence-initial temporal clauses, in accordance with the results observed in the previous studies. Regarding sentence-final temporal clauses, I expect to see the opposite situation - the "after"-clauses elicit more negative ERPs than "before"-clauses.

Regarding the second hypothesis:

(2) The percentage of the correctness of answers to the event- and temporal-order-related questions will be higher. Accordingly, the reaction time will be faster for the experimental trials (with colour-coding stimuli) than for the control trials (non-coloured stimuli).

I assume that this phenomenon will be due to the fact that there will be less demanding processing costs for the experimental (colour-coded) trials when it comes to perceiving and comprehending a sentence with temporal connectives.

This is achieved by creating a more evident distinction between event clauses and temporal order words. Therefore, the participants will pay more attention to the temporal connective used in the sentence. The colour-coded sentences are then perceived more from the perspective of a structure rather than from the perspective of a narrative. I believe this process could reduce both the time and the difficulty of mentally arranging events in a particular temporal order. Thus, the logical temporal order judgements are derived in less amount of time and with higher accuracy.

5 Pilot Study

The initial pilot study was conducted on a rather small amount of participants (5 University students; aged between 20-30 years old; right-handed; fluent English speakers). Neither EEG nor ERP recordings were not obtained. The experiment was conducted using only the experimental environment created on a Unity platform.

Obtained results were not notably significant without additional neuronal recordings data and deeper analysis. The test experiment trials have shown that the first trials within each block were causing a slower reaction time (10.2s on average across the first 5 trials for the first block; 4.1s on average across the first 5 trials for the second block) when compared to the reaction time obtained from the trials at the end of the block (1.06s on average for the last 5 trials of the first block; 2.32s - of the second block). I assume that these observations are due to the fact that participants develop behavioural patterns or strategies for recognizing and processing the temporal connections between the sentence structures and the temporal order of events. The progress at recognising colour-coded sentences at the beginning of a block did not differ significantly from the non-coloured sentences. Perhaps, it might be the case as the participants are not familiar with the sentence structures at the beginning of each block and are less skilled in recognising temporal patterns.

Comparing the efficiency of sentence perception and comprehension, as well as the quality of temporal order judgements, the results appeared to be controversial, based on the reaction time only. In the comparison of sentence-initial stimuli between "before"- and "after"-clauses, the average reaction time for the former compounds of 3.2s (sentence-initial 'before'), while for the latter - 6.33s (sentence-initial 'after'). In the case of sentence-final stimuli, the reaction time is 2,8s for "before"-sentences (sentence-final "before"), while for "after"-sentences - 4,1s (sentence-final "after"). These results vigorously contradict previously obtained findings from earlier studies. However, for more accurate results, more data is needed. Correctness remained sustainable (tendency towards correct answers) across all trials and conditions.

As for colour-coded vs. non-coloured sentences' reaction times, the average time is 5,27s and 3,13s accordingly (averaged over all types of sentence structures). These findings also do not support the initial hypothesis (2). I might assume, that the reason for that is visual colour stimulus being rather a distraction to the temporal order judgements rather than a helpful instrument. Nevertheless, at this stage of the research, it is difficult to make any exact conclusions on the matter of things in this regard. I believe that further improvements in the experiment design and procedure might be more substantial regarding deriving results.

6 Conclusion

Despite the fact that greatly significant results were not derived from the pilot study, as well as the number of participants was not sufficient, I believe that the experiment has more potential with additional data recordings and better settings.

I assume, in order to improve the experiment it would be beneficial to increase the sample size of participants. A larger sample size could potentially increase the statistical power of the experiment, allowing more precise and reliable results. Additionally, using a more diverse sample of participants could also help to generalize the results to a broader population, except the participants should still fit the experiment requirements (right-handed; English speakers; no mental or psychological disturbances).

As a recommendation for further implementation of the present experiment design, it may be useful to incorporate other measures of brain activity, such as functional magnetic resonance imaging (fMRI), to complement the ERP recordings and provide a more comprehensive understanding of the neural correlates of temporal order judgments.

Additionally, it may be beneficial to incorporate additional tasks or manipulations to better understand the cognitive processes involved in temporal order judgments. For example, manipulating the difficulty or complexity of the task, such as incorporating comprehension and filler questions. The comprehension questions could still relate to the temporal order judgments and question the causality of the presented events, while filler questions serve as a distraction from the main focus of the experiment, so the participants are aware to a less extent about the focus of the study and could not develop a behavioural pattern in response to the given stimuli.

Taking into consideration the use of source localization techniques to identify the brain regions involved in temporal order judgments, or frequency-domain analysis to examine the oscillatory dynamics of the brain during the task might also appear beneficial for the present study.

Overall, further developments within the present study could provide better insights into the neural mechanisms underlying temporal order judgments.

References

[1] Münte T. F., Schlitz, K., Kutas, M. (1998). When temporal terms belie conceptual order. In Nature, 395: pp. 71–73.

[2] Nieuwland, M. (2015). The Truth Before and After: Brain Potentials Reveal Automatic Activation of Event Knowledge During Sentence Comprehension. In Journal of Cognitive Neuroscience.

[3] Politzer-Ahles, S., Xiang, M., Almeida, D, (2017). "Before" and "after": Investigating the relationship between temporal connectives and chronological ordering using event-related potentials. In PloS one 12 (4).

[4] Mandler, J. (1986). On the comprehension of temporal order. In Lang Cog Proc. 1, pp. 309–320.

[5] Hendrich, E., Strobach, T., Buss. M., Müller, H., Schubert, T., (2012). Temporal-order judgment of visual and auditory stimuli: modulations in situations with and without stimulus discrimination. In Frontiers in Integrative Neuroscience. p. 63.

[6] Zwaan, R. A., Madden, C. J., Stanfield, R. A. (2001). Time in narrative comprehension. In D. H. Schram, G.J. Steen (Eds.), Psychology and Sociology of Literature, pp. 71–86.

[7] Jaśkowski P, Jaroszyk F, Hojan-Jezierska D. (1990). Temporal-order judgments and reaction time for stimuli of different modalities. Psychol Res. 1990; 52(1):35-8. doi: 10.1007/BF00867209. PMID: 2377723.

[8] Sereda, M., Szelag, E., Rojek, E., Pöppel, E. (2002). Temporal order judgement for auditory and visual stimuli. Acta neurobiologiae experimentalis. 62. 263-70.

[9] Constable, M. D., Welsh, T. N., Huffman, G., Pratt, J. (2019). I before U: Temporal order judgements reveal bias for self-owned objects. Quarterly Journal of Experimental Psychology, 72(3), 589–598. https://doi.org/10.1177/1747021818762010